Subject 2 : Prediction of selling price

BOUTON Hugo, DJOUSSA Chloé, GHAZALEH Wissam, MEHRIE Roxane

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1 Introduction

As part of this project, we are studying a dataset of 372 residential construction projects. The dataset contains 103 variables describing each project along several dimensions: temporal information (start and end year and quarter), physical characteristics (surface area, number of floor, lot area, etc.) and financial variables (estimated costs, etc.).

In addition to the technical characteristics specific to each project, the dataset also includes a set of macroeconomic variables and economic indices reflecting the overall context in which the projects were built. These variables include, for example, activity indicators such as the number of building permits issued, the total authorised floor area, and the Building Services Index (BSI). These indices allow for an analysis of the overall dynamics of the construction sector. They reflect both current activity and future prospects for the real estate market. They reflect the intentions of industry players, the upcoming workload, and trends in construction costs. By combining this information, it is possible to assess the health of the real estate market.

There are also price indices, such as the Wholesale Building Materials Price Index (WPI), the price of land, and several versions of the Consumer Price Index (CPI), including those specific to housing and energy. These indices help anticipate spending, adjust budgets, and assess project profitability while reducing risks associated with economic uncertainty. They also enable procurement strategies and resource management to be optimised, guaranteeing effective planning in a fluctuating economic environment.

Other variables provide information on the monetary and financial situation, such as cumulative liquidity, bank loans (number and amounts), interest rates, official and unofficial exchange rates, stock market indicators and the price of gold. These variables provide information on the availability of financing, the cost of credit, and international economic fluctuations. Stock market indicators and the price of gold reflect investor confidence, while all of this data helps assess the impact of monetary and financial conditions on the feasibility and

profitability of real estate projects.

The goal is to establish a model that can predict actual sales prices. It is therefore a supervised learning problem, and more precisely a regression problem.

The objective is therefore to find a function f, such that:

$$Y = f(X) + \epsilon$$

where Y corresponds to the target variable : "V-9", X represents the covariates*, and ϵ denotes a random error term.

*All variables except "V-9", "V-10", "START YEAR","START QUARTER","COMPLETION YEAR","COMPLETION QUARTER". We decide to remove "V-10" because it is predicted by another group. We remove also "START YEAR","START QUARTER","COMPLETION YEAR","COMPLETION QUARTER" because they do not degrade predictions much and we don't know how to interpret them.

Some variables in the dataset are associated with a field called "Time Lag Number," whose value is between 1 to 5. This means that these variables were measured or estimated at different times prior to the project's completion. This structure allows for the integration of a dynamic temporal dimension into the analysis. In our dataset, this translates into the presence of several versions of the same variable, each corresponding to a specific time lag. This mechanism allows us to capture the lagged effects that certain explanatory variables may have on the target variable.

2 Data pre-processing

Before applying certain models or statistical tests, it is important to check whether the variables under study follow a normal distribution. To do this, we applied three normality tests to the V-9 variable: the Shapiro-Wilk test, the D'Agostino-Pearson test and the Anderson-Darling test.

These tests share the null hypothesis that the data follow a normal distribution. The Shapiro-Wilk test is particularly suited to small samples and is sensitive to deviations in the extremities of the distribution. The D'Agostino-Pearson test combines measures of skewness and kurtosis, making it robust to medium to large sample sizes. Finally, the Anderson-Darling test provides a statistic to be compared with critical values for different levels of significance; it is often stricter than the first two.

The Shapiro-Wilk test gives a statistic of 0.8059 with a p-value strictly less than 0.0001, indicating rejection of the normality hypothesis. The D'Agostino-

Pearson test also produced a p-value of almost zero, confirming the existence of a strong deviation from the normal distribution, probably linked to marked asymmetry or kurtosis. Finally, the Anderson-Darling test was well above all critical values, even at the strictest significance level (1.081). All these results lead to the conclusion that the V-9 variable does not follow a normal distribution, which should be taken into account in the analysis, particularly when choosing the transformations or models to be applied.

3 Principal Component Analysis (PCA)

3.1 Introduction

Principal Component Analysis (PCA) is a statistical method used to study and simplify complex, multidimensional data sets. Its main objective is to reduce the number of variables to be analysed, while preserving the essential information contained in the initial data.

PCA involves transforming the original variables, which are often numerous and correlated, into a new set of variables called principal components. These components are constructed in such a way as to be independent of each other and to capture as much of the total variability in the data as possible. Thus, the first principal component explains most of the variance, the second explains the remaining maximum, and so on.

3.2 Mathematical theory

This method consists of finding a vector subspace E_k of dimension k such that the projected inertia J_{E_k} of the point cloud $N = \{x_i\}_{i \in I}$ in $\mathbb{R}^p$ with metric M is maximal. In other words, we are trying to project our data onto a smaller vector space (dimension reduction), while retaining a significant amount of information (maximisation of inertia).

Each point x_i is weighted by a mass $p_i > 0$ such that $\sum_{i \in I} p_i = 1$. Denoting G as the centroid of N and setting the origin at $O = G$, the problem reduces to diagonalising the matrix:

$$V_M = Y^\top D_p Y M,$$

where Y is the centred data matrix, and D_p is the diagonal matrix of weights.

The matrix V_M is M-symmetric and positive. It admits an M-orthonormal basis $(u_1, \ldots, u_p)$ of eigenvectors associated with decreasing real eigenvalues:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0.$$

The first k eigenvectors $(u_1, \ldots, u_k)$ then define the optimal subspace:

$$E_k = \text{Vect}(u_1, \ldots, u_k), \quad \text{with} \quad J_{E_k} = \sum_{i=1}^{k} \lambda_i.$$

The associated **principal components** are given by:

$$\Psi_\alpha = YMu_\alpha, \quad \text{for } \alpha = 1, \ldots, p.$$

Each Ψ_α is a centred variable with variance λ_α, and the principal components are pairwise uncorrelated. Thus, ψ_α is a column vector containing the coordinates of the individuals projected onto the factorial axis u_α.

The total inertia of the cloud is written as:

$$IT = \sum_{\alpha=1}^{p} \lambda_\alpha,$$

and the **explained inertia ratio** for the first k axes is:

$$\tau_{1,\ldots,k} = \frac{\lambda_1 + \cdots + \lambda_k}{IT}.$$

The explained inertia ratio by the first k principal components corresponds to the proportion of the total variance of the data captured by these k axes. It allows us to assess the amount of information retained after dimensionality reduction.

Usefulness of PCA in this project

In our project, the dataset contains over 100 explanatory variables, which complicates modelling and may introduce strong redundancy between variables. PCA makes it possible to:

- Reduce the dimensionality of the dataset without significant loss of information,
- Decorrelate the explanatory variables by constructing orthogonal principal components,
- Facilitate data visualisation and understanding of the underlying structure,
- Prepare a stable subspace to apply supervised learning methods (regression, etc.).

PCA is used here as a preprocessing step to reduce model complexity, improve prediction accuracy, and limit the risk of overfitting.

4 Variable selection methods and model construction

4.1 Variable selection using PCA

To identify the most relevant variables, we first carried out a principal component analysis. For each of the first four principal components (PC1 to PC4), we

analysed the projection coefficients (loadings) associated with the explanatory variables. These coefficients were sorted in decreasing order of their absolute value, which allows us to assess the importance of each variable in the construction of the principal axes.

We then selected the 8 most influential variables for PC1, the top 6 for PC2, and the top 2 for PC3 and PC4. The names of the selected variables were extracted and stored separately for each component. This approach aims to target the variables that contribute the most to the explained variance, facilitating interpretation of the factorial axes and dimensionality reduction.

4.2 Variable selection using F-Regression

In additon to PCA, we used the `f_regression` function from the `sklearn.feature_selection` module in order to valuate the relevance of individual features with respect to the target variable.

This method computes:

- the **F-statistic** (from the Fisher test) for each feature, which measures how strongly that feature is linearly related to the target, and
- the associated **p-value**, which assesses the statistical significance of the relationship.

This approach fits a separate univariate linear regression model for each feature, evaluating how well it alone can predict the target. Features with higher F-scores and low p-values are considered more relevant.

The F-statistic is calculated as:

$$F = \frac{\text{explained variance}}{\text{unexplained variance}} = \frac{(TSS - RSS)/p}{RSS/(n-p-1)},$$

where:

- TSS is the total sum of squares,
- RSS is the residual sum of squares,
- n is the number of observations,
- $p = 1$ since the regression is univariate.

To perform feature selection, we combined `f_regression` with `SelectKBest`, which retains the k features with the highest F-statistics (and correspondingly low p-values) which are considered more linearly correlated with the target variable. This technique helps in selecting the most relevant predictors and in our case it was the best statistical method to do it.

4.3 RMSE and R^2

To select the best model, we basically compared the R^2 coefficients and the RMSEs of the different models.

Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where:

- y_i is the actual (observed) value,
- $\hat{y}_i$ is the predicted value by the model,
- n is the total number of observations.

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{MSE}$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where:

- y_i is the actual (observed) value,
- $\hat{y}_i$ is the predicted value by the model,
- $\bar{y}$ is the mean of the observed values.

The numerator, $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, is the residual sum of squares (RSS), which measures the prediction error. The denominator, $\sum_{i=1}^{n}(y_i - \bar{y})^2$, is the total sum of squares (TSS), which measures the total variance in the data. Thus, R^2 represents the proportion of the variance in the dependent variable that is explained by the model.

4.4 Model choice and regularisation

We tested several types of models: linear regression, Poisson regression, and regularised models such as Ridge, Lasso, ElasticNet and LassoBIC (which automatically selects a regularisation parameter using the BIC criterion).

Linear regression is a common starting point, as it assumes a linear relationship between the target variable y and the explanatory variables X. However, the histogram of y shows a shape close to a Poisson distribution, which also justifies the use of a Poisson regression model, suited for discrete and strictly positive data.

To limit overfitting and improve generalisation, we used regularisation techniques. The Ridge model applies a penalty that shrinks the coefficients without setting them to zero, thus retaining all variables. Lasso, on the other hand, applies a different penalty that drives some coefficients to zero, simplifying the model by removing less useful variables.

When both approaches yield similar performance (in terms of R^2 and RMSE), Lasso is often preferred because it produces simpler and more interpretable models, while maintaining good predictive performance (even if in this project, we focus on prediction and not explanation).

Moreover, we tested KNN and Neural Networks. K-nearest neighbors (KNN) is a non-parametric regression method that predicts the target value for a given input based on the average of the target values of the k nearest training samples in feature space. The method is simple and intuitive, making no assumptions about the data distribution hence our choice. However, KNN's performance heavily depends on the choice of k, the distance metric, and the scaling of features. It can be computationally expensive for large datasets since predictions require calculating distances to all training samples. Furthermore, KNN may struggle in high-dimensional spaces due to the "curse of dimensionality," which can dilute the significance of proximity. Neural Networks is described in the following sections.

4.5 Result variability and model robustness

To make our results reproducible, we used different random seeds (such as 5 and 47) to split the data into training and test sets. However, despite these precautions, we observed that model performance varies depending on the chosen seed. A model may perform best with one data split, but be outperformed by another in a different split.

This variation in results shows that model performance is highly dependent on the random draw of the data. It is therefore difficult to state definitively which model is truly the best. To address this variability and obtain a more reliable evaluation, it is advisable to use methods such as cross-validation, which test models across multiple independent splits of the dataset.

To conclude, we compare two methods and present the results of our study:

5 ElasticNet method

ElasticNet is a linear regression model that combines both L1 (Lasso) and L2 (Ridge) regularization penalties. This combination allows the model to perform effective feature selection (via L1) while also stabilizing the coefficient estimates in the presence of multicollinearity (via L2). ElasticNet is particularly useful when working with high-dimensional datasets or when predictor variables are highly correlated.

In our project, the dataset contained many correlated features. To address this problem and reduce the risk of overfitting, we selected ElasticNet regression as one of our main modeling approaches.

To tune ElasticNet regression model, we used cross-validation to select the optimal values for the regularization parameters: alpha, which controls the overall strength of the regularization, and $L1$ *ratio*, which determines the balance between L1 (Lasso) and L2 (Ridge) penalties. This process allowed the model to select a subset of important features (those with non-zero coefficients) and helped prevent overfitting, ensuring good performance on the test set.

We finally chose the ElasticNet method to build our final model, because it gave the best results in terms of R^2 and RMSE. In addition to its good performance, it has the advantage of being less sensitive to the random selection of data (seed), unlike methods such as PCA, whose results can vary from one run to the next. ElasticNet is therefore more stable. Another interesting point is that we still have access to the coefficients of the model, which allows us to better understand the influence of each variable, even if, in this case, interpretation was not our main objective.

6 Artificial Neural Network method

Artificial Neural Network are more complex than linear models. Indeed, Neural network models mimic the way our brain processes information by passing data through layers of interconnected nodes, each one learning to recognize different patterns. As the network adjusts its internal weights during training, it gradually captures complex, non-linear relationships, enabling it to make increasingly accurate predictions. It's a robust predictor when we provide the model a carefully chosen range of features. In our case, we used GridSearchCV to make sure that the hyperparameters (such as the number of layers, the number of neurons, activation function, ...) are well-tuned to the data at hand, and the cross-validation and learning curve evaluations further reinforces the model's generalization capabilities. The reason we did this step is to avoid overfitting or underfitting.

We can say that overall, the artificial neural network model is a strong predictor, however the neural network is harder to explain due to its non-linear and "black-box" nature, as we actually don't know the value of the coefficients. Moreover, its prediction is less effective compared to the ElasticNet (based on R^2 and RMSE) as well as its "lack" of interpretability, which is why we will keep the ElasticNet Model.